heat before taking off. Once flying, the muscles use large amounts of chemical energy but only some of it is used to beat the wings; the rest becomes more heat. This, combined with the warmth of direct sunlight, means a flying insect risks overheating.

To solve this problem, many species have a heat-exchange mechanism that shifts excess heat from the thorax to the abdomen, allowing the insect to maintain a steady temperature in its thorax.

Range of techniques

By changing the angle of their wings, butterflies control their body temperature. When they are trying to warm up, holding their wings wide open maximizes the amount of sunshine falling on them. When they are trying to cool down, they move into shade or angle their wings upward so that less direct sunlight shines on their surface.

Other insects use even more remarkable methods to regulate their body temperature. When a mosquito drinks the warm blood of a mammal, this raises its body temperature. To compensate, it produces droplets of fluid that are kept at the end of the abdomen; evaporative cooling of these droplets lowers the insect’s temperature. Dung beetles construct balls of dung in which females lay their eggs. Some dung beetles are able to raise the temperature of their thorax so they can roll heavier balls.

The range of thermoregulation techniques shows how life forms evolve to better fit their environment. They can also inspire technology: arrays of solar panels angled to track the Sun capture maximum amounts of solar radiation—just like butterfly wings.